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FIRST ORDER QUASILINEAR EQUATIONS  
WITH BOUNDARY CONDITIONS

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ABSTRACT : We solve the initial and boundary condition problem for a general first order quasilinear equation in several space variables by using a vanishing viscosity method and give a definition which characterizes the obtained solution.

Let  $\Omega$  be a bounded set of  $\mathbb{R}^p$  with a piecewise regular boundary  $\Gamma$  of exterior normal unit vector  $n$ , and a real number  $T > 0$ . We consider the problem of finding a unique solution (in a sense to be defined later)  $u$  of the first order quasilinear equation :

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$$(1) \quad \frac{\partial u}{\partial t} + \sum_{i=1}^p \frac{\partial}{\partial x_i} [f_i(u, x, t)] + g(u, x, t) = 0$$

on  $\Omega \times ]0, T[$  which satisfies a given initial condition

$$(2) \quad u(x, 0) = u_0(x) \quad \text{for } x \in \Omega,$$

and a boundary condition

$$(3) \quad u(x, t) = a(x, t)$$

on a part of  $\Gamma \times ]0, T[$  to be specified later.

We suppose that  $f = (f_1, \dots, f_p)$ ,  $g$  and the data satisfy the following hypotheses

$$(4) \quad \begin{cases} f \in \left\{ C^2(\mathbb{R} \times \bar{\Omega} \times ]0, T[) \right\}^p, \\ g \in C^2(\mathbb{R} \times \bar{\Omega} \times ]0, T[), \\ u \in C^2(\bar{\Omega}), \\ a \in C^2(\Gamma \times ]0, T[). \end{cases}$$

Moreover,  $g$  and each  $\frac{\partial f_i}{\partial x_i}$  are assumed to be Lipschitz continuous with respect to  $u$ , uniformly in  $(x, t) \in \Omega \times ]0, T[$ . This last hypothesis ensures that the solution to be found is bounded, and then  $f$  and  $g$  need only be defined on a compact set. We can remark, by introducing an extension  $\bar{a} \in C^2(\bar{\Omega} \times ]0, T[)$  of the boundary data  $a$ , that equation (1) may be written in the new unknown function  $v = u - \bar{a}$ . Thus we obtain a new problem with a homogeneous boundary condition (i.e.  $a = 0$ ).

We need some properties of functions which belong to the space  $BV(\Omega \times ]0, T[)$ , of functions of bounded variation in the sense of

Tonelli-Cesari. The following lemma deals with the trace of such functions on the boundary.

LEMMA 1 : For  $u \in BV(\Omega \times ]0, T[)$ , we can define a trace  $\gamma u$  in  $L^\infty(\Omega)$  for  $t=0$  and in  $L^\infty(\Gamma \times ]0, T[)$ , which is reached through  $L^1$  convergence. Moreover, for  $h \in C^1(\mathbb{R})$ , we have

$$\gamma [h(u)] = h [\gamma(u)],$$

almost everywhere on  $\Gamma \times ]0, T[$ .

Proof of Lemma 1 : Since  $u$  belongs to  $BV(\Omega \times ]0, T[)$ , for almost every  $x \in \Omega$ , the function  $u(x, \cdot)$  belongs to  $BV(0, T)$  and, then it has a left and right limit at each  $t \in ]0, T[$ , and a right limit for  $t=0$ . Thus we can define  $u(x)$  as the limit of  $u(x, t)$ , when  $t > 0$  tends to zero. Now, from Lebesgue's theorem, since  $u$  is uniformly bounded on  $\Omega \times ]0, T[$ , we obtain  $u \in L^\infty(\Omega)$  and  $u(\cdot, t)$  tends to  $\gamma u$  in  $L^1(\Omega)$  when  $t$  tends to zero.

Denoting by  $(e_1, \dots, e_p)$  the canonical base of  $\mathbb{R}^p$ , we can define with the same arguments as above a trace  $\gamma_i u$  at each point  $(x, t)$  of  $\Gamma \times ]0, T[$ , which is obtained by passing to the limit along the direction  $e_i$  for fixed  $t \in ]0, T[$  and for all  $i \in \{1, \dots, p\}$  for which  $e_i$  is not tangent to the boundary at point  $(x, t)$ . Now we introduce  $du(x, \cdot)$ , the Radon measure associated with the distribution  $\text{grad } u(x, \cdot) = (\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_p})$  (this is a well-known property of space  $BV$ ), and we obtain, for all  $\phi \in (C^1(\overline{\Omega}))^p$  and  $i \in \{1, \dots, p\}$  (for which  $e_i$  is not tangent to the boundary on the support of  $\phi$ )

$$(5) \quad \int_{\Gamma} \gamma_i u n_i \phi \, ds = \int_{\Omega} u \operatorname{div} \phi \, dx + \int_{\Omega} \phi \cdot du(x, \cdot),$$

where the scalar product on  $\mathbf{R}$  is denoted by a dot. From (5), it is obvious that  $\gamma_i u = \gamma_j u$  almost everywhere on  $\Gamma \times ]0, T[$ . That allows us to define a unique trace  $\gamma u \in L^\infty(\Gamma \times ]0, T[)$  which is reached by  $L^1$  convergence. The last step of the lemma follows from this  $L^1$  convergence of  $u$  towards its trace, and thus Lemma 1 is proved. ■

We also recall the following lemma, from SAKS [9, pp. 130-131].

LEMMA 2 : If  $v \in C^1(\overline{\Omega})$ , then

$$\lim_{\eta \rightarrow 0} \int_{\{x \in \Omega; |v(x)| < \eta\}} |\text{grad } v| \, dx = 0.$$

We now introduce the sign function  $sg$  and its approximation  $sg_\eta$  for  $\eta > 0$ , defined on  $\mathbf{R}$  by

$$sg(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} ; \quad sg_\eta(x) = \begin{cases} 1 & \text{if } x > \eta \\ x/\eta & \text{if } |x| \leq \eta \\ -1 & \text{if } x < -\eta \end{cases}.$$

For  $\xi \in \mathbf{R}^p$ , we can write

$$|\xi| = (\xi_1^2 + \dots + \xi_p^2)^{\frac{1}{2}},$$

and then define  $I_\eta$  on  $\mathbf{R}^p$  by

$$I_\eta(\xi) = \int_0^{|\xi|} sg_\eta(y) \, dy.$$

We have, for all  $i \in \{1, \dots, p\}$ ,

$$\frac{\partial I_\eta}{\partial \xi_i} = sg_\eta(|\xi|) \frac{\xi_i}{|\xi|},$$

and we show easily that  $I_\eta$  is a convex function on  $\mathbb{R}^p$ .

To prove existence, we use the vanishing viscosity method which consists in passing to the limit, as  $\varepsilon > 0$  tends to zero, on the unique solution  $u_\varepsilon \in C^2(\bar{\Omega} \times ]0, T[)$  of the quasilinear parabolic equation

$$(6) \quad \frac{\partial u_\varepsilon}{\partial t} + \operatorname{div} [f(u_\varepsilon, x, t)] + g(u_\varepsilon, x, t) = \varepsilon \Delta u_\varepsilon$$

on  $\Omega \times ]0, T[$ , with the following initial and boundary conditions :

$$(7) \quad u_\varepsilon(x, 0) = u_0(x) \quad \text{for } x \in \Omega,$$

$$(8) \quad u_\varepsilon(x, t) = 0 \quad \text{for } (x, t) \in \Gamma \times ]0, T[.$$

The right-hand side in (6) is a regularizing term, which physically corresponds to a viscosity of parameter  $\varepsilon$ , since this problem is sometimes used as a model for fluid motions.

Existence and uniqueness for problem (6), (7), (8), is known (see, for instance, A. FRIEDMAN [2] or O.A. LADYZENSKAYA-N.N. URAL'CEVA [4]) and we study here the convergence of  $\{u_\varepsilon\}_{\varepsilon > 0}$  as  $\varepsilon$  tends to zero via a compactness argument in  $L^1(\Omega \times ]0, T[)$ . This argument ensures also that a limit  $u$  of a sequence  $\{u_{\varepsilon_m}\}_m$  belongs to  $BV(\Omega \times ]0, T[)$ . As an argument for uniqueness, we shall afterwards state that  $u$  is the solution of problem (1), (2), with the prescribed boundary condition which satisfies a so-called entropy condition. As a first step, we obtain convergence by the following theorem.

**THEOREM 1 :** *The family  $\{u_\varepsilon\}_{\varepsilon > 0}$  of the solutions of problem (6), (7), (8), is sequentially compact in  $L^1(\Omega \times ]0, T[)$  and each limit function  $u$  belongs to  $BV(\Omega \times ]0, T[)$  and has a trace for  $t=0$  satisfying (2).*

Proof of Theorem 1 : We shall estimate  $u_\varepsilon$  in the Sobolev space  $W^{1,1}(\Omega \times ]0,T[)$ . First, from the maximum principle (see, for example [2]), we obtain

$$(9) \quad |u_\varepsilon|_{L^\infty(\Omega \times ]0,T[)} \leq |u_0|_{L^\infty(\Omega)} e^{C_0 T},$$

where  $C_0$  is the sum of the Lipschitz constants of  $g$  and  $\frac{\partial f_i}{\partial x_i}$  for all  $i \in \{1, \dots, p\}$ , with respect to the variable  $u$ .

Let us now take the derivative of equation (6) with respect to  $t$ , multiply by  $sg_\eta(\frac{\partial u_\varepsilon}{\partial t})$  for  $\eta > 0$  and integrate over  $\Omega$  to obtain

$$(10) \quad \left\{ \begin{aligned} & \frac{d}{dt} \int_{\Omega} \left[ \int_0^{\frac{\partial u_\varepsilon}{\partial t}} sg_\eta(v) dv \right] dx + \int_{\Omega} \operatorname{div} \left[ \frac{\partial f}{\partial u}(u_\varepsilon, x, t) \frac{\partial u_\varepsilon}{\partial t} \right] sg_\eta \left( \frac{\partial u_\varepsilon}{\partial t} \right) dx \\ & + \int_{\Omega} \left\{ \operatorname{div} \left( \frac{\partial f}{\partial t}(u_\varepsilon, x, t) \right) + \frac{\partial}{\partial t} g(u_\varepsilon, x, t) \right\} \\ & \quad \quad \quad sg_\eta \left( \frac{\partial u_\varepsilon}{\partial t} \right) dx \\ & = \varepsilon \int_{\Omega} \Delta \left( \frac{\partial u_\varepsilon}{\partial t} \right) sg_\eta \left( \frac{\partial u_\varepsilon}{\partial t} \right) dx. \end{aligned} \right.$$

We integrate by parts the second integral and use the homogeneous boundary condition to get this term equal to

$$- \int_{\Omega} \frac{\partial f}{\partial u}(u_\varepsilon, x, t) \cdot \operatorname{grad} \left( \frac{\partial u_\varepsilon}{\partial t} \right) \frac{\partial u_\varepsilon}{\partial t} sg'_\eta \left( \frac{\partial u_\varepsilon}{\partial t} \right) dx,$$

which tends to zero with  $\eta$  by Lemma 2. In the same way, the right-hand side yields

$$- \varepsilon \int_{\Omega} |\operatorname{grad} \left( \frac{\partial u_\varepsilon}{\partial t} \right)|^2 sg'_\eta \left( \frac{\partial u_\varepsilon}{\partial t} \right) dx,$$

which is never positive. The remaining term (i.e. the second line of (10)) may be estimated by the  $L^1$ -norm of the first derivatives of  $u_\varepsilon$ ,

and thus we obtain in the limit, when  $\eta$  tends to zero, for all  $t > 0$ ,

$$(11) \quad \left\{ \begin{aligned} \left| \frac{\partial u_\epsilon}{\partial t}(\cdot, t) \right|_{L^1(\Omega)} &\leq \left| \frac{\partial u_\epsilon}{\partial t}(\cdot, 0) \right|_{L^1(\Omega)} + \int_0^t \left\{ c_1 + c_2 \left( \left| \frac{\partial u_\epsilon}{\partial t}(\cdot, s) \right|_{L^1(\Omega)} \right. \right. \\ &\quad \left. \left. + \left| \text{grad } u_\epsilon(\cdot, s) \right|_{L^1(\Omega)} \right) ds \right\} \end{aligned} \right.$$

where  $c_1$  and  $c_2$  are constants which do not depend on  $\epsilon$ .

For  $\epsilon$  bounded (which can be assumed), we obtain from equation (6):

$$(12) \quad \left| \frac{\partial u_\epsilon}{\partial t}(\cdot, 0) \right|_{L^1(\Omega)} \leq c + c' \|u_0\|_{W^{2,1}(\Omega)},$$

where  $c$  and  $c'$  are constants which depend only on  $g$  and  $f$ .

Now we try to obtain the same estimates as (11), (12), for  $\text{grad } u_\epsilon$ . Let us take the gradient of both sides of (6) and get the scalar product by the vector which components are

$$\frac{\partial I_\eta}{\partial \xi_j}(\text{grad } u_\epsilon), \quad 1 \leq j \leq p,$$

and integrate over  $\Omega$  to obtain, after some integrations by part,

$$(13) \quad \left\{ \begin{aligned} \frac{d}{dt} \int_\Omega I_\eta(\text{grad } u_\epsilon) dx &= \\ &= - \int_\Omega \sum_i \sum_j \sum_k \frac{\partial^2 I_\eta}{\partial \xi_j \partial \xi_k}(\text{grad } u_\epsilon) \frac{\partial^2 u_\epsilon}{\partial x_i \partial x_j} \frac{\partial^2 u_\epsilon}{\partial x_i \partial x_k} dx \\ &\quad - \int_\Omega \text{div} \left( \frac{\partial f}{\partial u}(u_\epsilon, x, t) \right) \left\{ |\text{grad } u_\epsilon| \text{sg}_\eta(|\text{grad } u_\epsilon|) - I_\eta(\text{grad } u_\epsilon) \right\} dx \\ &\quad - \sum_j \int_\Omega \left\{ \frac{\partial}{\partial x_j} (g(u_\epsilon, x, t)) + \text{div} \left( \frac{\partial f}{\partial x_j}(u_\epsilon, x, t) \right) \right\} \frac{\partial u_\epsilon}{\partial x_j} \frac{\text{sg}_\eta(|\text{grad } u_\epsilon|)}{|\text{grad } u_\epsilon|} dx \\ &\quad + \int_\Gamma \left\{ \epsilon \frac{\partial}{\partial n} \left( I_\eta \left( \frac{\partial u_\epsilon}{\partial n} n \right) \right) - \frac{\partial f}{\partial u}(0, s, t) \cdot n I_\eta \left( \frac{\partial u_\epsilon}{\partial n} n \right) \right\} ds. \end{aligned} \right.$$



We now consider each term of the right-hand side of (13). Since  $I_\eta$  is a convex function, the first term is negative. The second term tends to zero with  $\eta$ , and the third one may be estimated by means of the  $L^1$ -norm of the first space derivatives of  $u_\varepsilon$ , uniformly with respect to  $\eta$ . Thus there remains the integral on the boundary in which we introduce equation (6) rewritten as follows, for all  $t \in ]0, T[$  and at a point  $x$  of the boundary :

$$\frac{\partial f}{\partial u}(0, x, t) \cdot n \frac{\partial u_\varepsilon}{\partial n} = \varepsilon \Delta u_\varepsilon - g(0, x, t) - \sum_i \frac{\partial f_i}{\partial x_i}(0, x, t).$$

Thus we get the last term of (13) equal to

$$\left\{ \begin{aligned} & \varepsilon \int_\Gamma \left\{ \frac{\partial^2 u_\varepsilon}{\partial n^2} \operatorname{sg}_\eta \left( \frac{\partial u_\varepsilon}{\partial n} \right) - \varepsilon \Delta u_\varepsilon \frac{I_\eta \left( \frac{\partial u_\varepsilon}{\partial n} n \right)}{\frac{\partial u_\varepsilon}{\partial n}} \right\} ds \\ & + \int_\Gamma \left\{ g(0, x, t) + \sum_i \frac{\partial f_i}{\partial x_i}(0, x, t) \right\} \frac{I_\eta \left( \frac{\partial u_\varepsilon}{\partial n} n \right)}{\frac{\partial u_\varepsilon}{\partial n}} ds. \end{aligned} \right.$$

The second term of this sum is uniformly bounded with respect to  $\varepsilon$  and  $\eta$ , while, as  $\eta$  tends to zero, the limit of the first term can be estimated by

$$\varepsilon \int_\Gamma \left| \frac{\partial^2 u_\varepsilon}{\partial n^2} - \Delta u_\varepsilon \right| ds.$$

Now, by a local computation (i.e. we write  $\Delta u_\varepsilon$  in tangential and normal coordinates), we find a constant  $c_3$  depending on the curvature of  $\Gamma$  so that this last integral is bounded by

$$\varepsilon c_3 \int_\Gamma \left| \frac{\partial u_\varepsilon}{\partial n} \right| ds.$$

From [1], we have

$$\left| \frac{\partial u_\varepsilon}{\partial n} \right|_{L^1(\Omega)} \leq |\Delta u_\varepsilon|_{L^1(\Omega)},$$

since  $u_\varepsilon$  is zero on the boundary. By using equation (6) to estimate  $\varepsilon c_3 \|\Delta u_\varepsilon\|_{L^1(\Omega)}$  by means of the  $L^1$ -norm of the first derivatives of  $u_\varepsilon$ , we obtain the estimation of the second member of (13) and we can state that, for all  $t \in ]0, T[$ ,

$$(14) \quad \left\{ \begin{aligned} \|\text{grad } u_\varepsilon(\cdot, t)\|_{L^1(\Omega)} &\leq \|u_0\|_{W^{1,1}(\Omega)} + c_4 t + c_5 \int_0^t \left\{ \|\text{grad } u_\varepsilon(\cdot, s)\|_{L^1(\Omega)} \right. \\ &\quad \left. + \left\| \frac{\partial u_\varepsilon}{\partial t}(\cdot, s) \right\|_{L^1(\Omega)} \right\} ds. \end{aligned} \right.$$

By using Gronwall's lemma, we conclude from (9), (11), (12) and (14), that  $u_\varepsilon$  is uniformly bounded in  $W^{1,1}(\Omega \times ]0, T[)$ . We have

$$(15) \quad \|u_\varepsilon\|_{L^\infty(\Omega \times ]0, T[)} + \|\text{grad } u_\varepsilon\|_{L^1(\Omega \times ]0, T[)} + \left\| \frac{\partial u_\varepsilon}{\partial t} \right\|_{L^1(\Omega \times ]0, T[)} \leq K_0$$

for a constant  $K_0$  which does not depend on  $\varepsilon$ .

By the Riesz-Tamarkin theorem, the family  $\{u_\varepsilon\}_{\varepsilon > 0}$  is sequentially compact in  $L^1(\Omega \times ]0, T[)$ . The same estimate (15) allows us to claim that  $u_\varepsilon$  is uniformly bounded in  $BV(\Omega \times ]0, T[)$ , and then a limit  $u$  of a  $L^1$  converging sequence  $\{u_{\varepsilon_m}\}_m$  belongs to  $BV(\Omega \times ]0, T[)$ .

Such a limit  $u$  has a trace for  $t=0$  which is, by Lemma 1, the  $L^1(\Omega)$  limit of  $u(\cdot, t)$  when  $t$  tends to zero; this trace will be denoted by  $u(\cdot, 0)$ . By the Fubini theorem and Lebesgue's theorem, for almost all  $t \in ]0, T[$ , a subsequence of  $\{u_{\varepsilon_m}(\cdot, t)\}_m$  converges towards  $u(\cdot, t)$  in  $L^1(\Omega)$ , and then, from the inequality

$$\left\{ \begin{aligned} \|u(\cdot, 0) - u_0\|_{L^1(\Omega)} &\leq \|u(\cdot, 0) - u(\cdot, t)\|_{L^1(\Omega)} + \|u(\cdot, t) - u_\varepsilon(\cdot, t)\|_{L^1(\Omega)} \\ &\quad + \|u_\varepsilon(\cdot, t) - u_0\|_{L^1(\Omega)}, \end{aligned} \right.$$

and in view of

$$\|u(\cdot, t) - u_0\|_{L^1(\Omega)} \leq K_0 t,$$

which follows from (15), we conclude  $u(\cdot, 0)$  is equal to  $u_0$  almost everywhere on  $\Omega$  by passing to the limits on  $\epsilon$  and then on  $t$ . Thus Theorem 1 is proved. ■

Let  $u \in BV(\Omega \times ]0, T[)$  be the  $L^1$  limit of a sequence  $\{u_{\epsilon_m}\}_m$ ; from Theorem 1,  $u$  exists and satisfies (2), but the homogeneous boundary condition (8) on  $u_\epsilon$  can be no longer verified by  $u$ , as shown by the following example.

For Burgers equation on  $]0, 1[$ ,

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left( \frac{u^2}{2} \right) = 0,$$

with the initial value

$$u(x, 0) = 1,$$

we find that the exact solution (given by the vanishing viscosity method) for  $t < 1$  is

$$u(x, t) = \begin{cases} x/t & \text{for } x < t, \\ 1 & \text{for } t < x. \end{cases}$$

Obviously, the homogeneous boundary condition is not satisfied at  $x = 1$ , and this corresponds to a boundary layer on  $u_\epsilon$ . That can be easily explained by considering the propagation of the data, the speed of which is given by the characteristics equation, which is

$$\frac{\partial x}{\partial t} = u,$$

for the Burgers equation. The data are leaving  $\Omega$  at  $x = 1$ , while

they stay in  $\Omega$  as a rarefaction wave at  $x=0$ . For the linear problem on  $]0,1[$ , with  $\lambda \in \mathbb{R}$ ,

$$\frac{\partial u}{\partial t} + \lambda \frac{\partial u}{\partial x} = 0 ,$$

we find that the data have a constant speed, equal to  $\lambda$ , and then we can take into account a boundary condition at  $x=0$  in the case  $\lambda > 0$  but not for  $\lambda \leq 0$ . We have the reverse at  $x=1$ . In the general case, the correct way to write the boundary condition is the following :

$$(16) \quad \min_{k \in I(\gamma u, 0)} \left\{ \text{sg}(\gamma u) (f(\gamma u, x, t) - f(k, x, t)) \cdot n \right\} = 0 ,$$

where  $\gamma u$  is the trace of  $u$  on  $\Gamma \times ]0, T[$ , and  $I(\gamma u, 0)$  denotes the closed interval  $[\text{Min}(\gamma u, 0), \text{Max}(\gamma u, 0)]$ . For a non homogeneous boundary condition like (3), we must write instead of (16) :

$$(16.a) \quad \min_{k \in I(\gamma u, a)} \left\{ \text{sg}(\gamma u - a) (f(\gamma u, x, t) - f(k, x, t)) \cdot n \right\} = 0 ,$$

at a point  $(x, t) \in \Gamma \times ]0, T[$ . This minimum is ensured for  $k = u$ , when the data are leaving  $\Omega$ , and then the boundary condition is not necessarily satisfied. For entering data, we obtain  $\gamma u = a$ , which obviously verifies (16.a). In the linear case  $(f(u, x, t) = \lambda u)$  on  $\Omega = ]0, 1[$ , condition (16.a) becomes at  $x=1$  :

$$\min_{k \in I(\gamma u, a)} \left\{ \lambda |\gamma u - k| \right\} = 0 .$$

It is satisfied at  $k = \gamma u$  for  $\lambda \geq 0$ , independently from the prescribed boundary condition  $a$ , while, for  $\lambda < 0$ , this minimum can be zero only when  $\gamma u$  is equal to  $a$ .

Now, problem (1), (2), (16) is a well-posed problem. It remains to prove that  $u$  is its unique solution. Our definition of solution generalizes the definition proposed by S.N. KRUSKOV [3] for the problem on  $\mathbb{R}^p$  (without any boundary condition), and includes a so-called entropy condition that characterizes the solution given by the vanishing viscosity method. This solution is the only one with a relevant physical meaning when discontinuities appear. See [5], [8] for more details.

**DEFINITION** : A function  $u \in BV(\Omega \times ]0, T[)$  is a solution of problem (1), (2), (16), if, for all  $k \in \mathbb{R}$  and all non negative test function  $\phi \in C^2(\bar{\Omega} \times ]0, T[)$  with a compact support in  $\bar{\Omega} \times ]0, T[$ , it satisfies the following inequality :

$$(17) \quad \left\{ \begin{array}{l} \int_{\Omega} \int_0^T \left\{ |u-k| \frac{\partial \phi}{\partial t} + \text{sg}(u-k)(f(u, x, t) - f(k, x, t)) \cdot \text{grad } \phi \right. \\ \quad \left. - \left[ \sum_1 \frac{\partial f_i}{\partial x_1}(k, x, t) + g(u, x, t) \right] \text{sg}(u-k) \phi \right\} dx dt \\ \quad + \int_{\Omega} \int_0^T \text{sg}(k) \left\{ f(\gamma u, s, t) - f(k, s, t) \right\} \cdot n \phi ds dt \geq 0, \end{array} \right.$$

and the initial condition (2) almost everywhere on  $\Omega$ .

**THEOREM 2** : Problem (1), (2), (16), admits a unique solution which is the one given by the vanishing viscosity method.

**Proof of Theorem 2** : Let  $u$  be the  $L^1$  limit of a converging sequence  $\{u_{\varepsilon_m}\}_m$  of solutions of problem (6), (7), (8). By Theorem 1,  $u$  satisfies (2). To obtain (17), we introduce  $k \in \mathbb{R}$  and a test function

$\phi \geq 0$  as prescribed in the above definition, and we multiply equation (6) by  $sg_\eta(u_\varepsilon - k)\phi$  for  $\eta > 0$  and  $\varepsilon \in \{\varepsilon_m\}_m$ . After integrating by parts, we obtain

$$(18) \quad \left\{ \begin{aligned} & \int_{\Omega} \int_0^T \left\{ \int_k^{u_\varepsilon} sg_\eta(v-k) dv \right\} \frac{\partial \phi}{\partial t} dx dt \\ & + \int_{\Omega} \int_0^T sg_\eta(u_\varepsilon - k) (f(u_\varepsilon, x, t) - f(k, x, t)) \cdot \text{grad } \phi dx dt \\ & - \int_{\Omega} \int_0^T sg_\eta(u_\varepsilon - k) \left\{ \sum_i \frac{\partial f_i}{\partial x_i}(k, x, t) + g(u_\varepsilon, x, t) \right\} \phi dx dt \\ & + \int_{\Omega} \int_0^T (f(u_\varepsilon, x, t) - f(k, x, t)) \cdot \text{grad } u_\varepsilon sg'_\eta(u_\varepsilon - k) \phi dx dt \\ & = \varepsilon \int_{\Omega} \int_0^T sg'_\eta(u_\varepsilon - k) |\text{grad } u_\varepsilon|^2 \phi dx dt \\ & + \varepsilon \int_{\Omega} \int_0^T sg_\eta(u_\varepsilon - k) \text{grad } u_\varepsilon \cdot \text{grad } \phi dx dt \\ & + \varepsilon \int_{\Omega} \int_0^T sg_\eta(k) \phi \frac{\partial u_\varepsilon}{\partial n} ds dt \\ & - \int_{\Omega} \int_0^T sg_\eta(k) (f(0, s, t) - f(k, s, t)) \cdot n \phi ds dt. \end{aligned} \right.$$

Now let  $\eta$  tend to zero; the first three terms of (18) give the first member of (17) with  $u$  instead of  $u_\varepsilon$ , and the fourth term tends to zero by Lemma 2. The first term of the second member of (18) is non negative and the  $\eta$ -limit is easily obtained on each of the remaining terms. We now make  $\varepsilon$  tend to zero on the derived equality. It is easy to pass to the limit in the first member and that gives the first term in (17). By (15), we have

$$(19) \quad \left| \varepsilon \int_{\Omega} \int_0^T sg(u - k) \text{grad } u_\varepsilon \cdot \text{grad } \phi dx dt \right| \leq \varepsilon K_0 |\text{grad } \phi|_{L^\infty}$$

which tends to zero with  $\varepsilon$ . In order to get the limit on the

following term, we compute

$$\lim_{\varepsilon \rightarrow 0} \left\{ \varepsilon \int_{\Gamma} \int_0^T \phi \frac{\partial u_{\varepsilon}}{\partial n} ds dt \right\},$$

by using equation (6). Thus, for  $\delta > 0$ , we introduce a function

$\rho_{\delta} \in C^2(\bar{\Omega})$  verifying the following properties (see S.N. KRUSKOV [3])

$$\begin{cases} \rho_{\delta} \equiv 1 & \text{on } \Omega; \\ \rho_{\delta} \equiv 0 & \text{on } \{x \in \Omega; \text{dist}(x, \Gamma) \geq \delta\}; \\ 0 \leq \rho_{\delta} \leq 1 & \text{on } \Omega; \\ |\text{grad } \rho_{\delta}|_{L^{\infty}(\Omega)} \leq c/\delta, \end{cases}$$

where  $c$  is a constant independent of  $\delta$ . We have

$$\int_{\Gamma} \int_0^T \varepsilon \frac{\partial u_{\varepsilon}}{\partial n} \phi ds dt = \int_{\Omega} \int_0^T \varepsilon \left\{ \Delta u_{\varepsilon} \phi \rho_{\delta} + \text{grad } u_{\varepsilon} \cdot \text{grad}(\phi \rho_{\delta}) \right\} dx dt.$$

By using (19) and equation (6), we obtain after integrating by parts:

$$\begin{cases} \lim_{\varepsilon \rightarrow 0} \int_{\Omega} \int_0^T \varepsilon \frac{\partial u_{\varepsilon}}{\partial n} \phi ds dt = \\ = \int_{\Omega} \int_0^T \left\{ u \frac{\partial \phi}{\partial t} + f(u, x, t) \cdot \text{grad } \phi - g(u, x, t) \phi \right\} \rho_{\delta} dx dt \\ - \int_{\Omega} \int_0^T \phi f(u, x, t) \cdot \text{grad } \rho_{\delta} dx dt \\ + \int_{\Omega} \int_0^T \phi f(0, s, t) \cdot n ds dt. \end{cases}$$

Now, let  $\delta$  tend to zero. The left-hand side remains unchanged while the first term of the right-hand side tends to zero and the second term gives an integral over  $\Gamma$ . By using a Green's formula similar to (5) and Lebesgue's theorem, we obtain

$$\lim_{\varepsilon \rightarrow 0} \int_{\Gamma} \int_0^T \varepsilon \frac{\partial u_{\varepsilon}}{\partial n} \phi ds dt = \int_{\Gamma} \int_0^T \phi (f(0, s, t) - f(\gamma u, s, t)) \cdot n ds dt.$$

The  $\varepsilon$ -limit on (18) is now performed, since the last term does not depend on  $\varepsilon$ . We may remark that the term

$$\int_{\Gamma} \int_0^T \operatorname{sg}(k) f(0, s, t) \cdot n \phi \, ds \, dt$$

is eliminated and thus (17) is shown.

We now prove uniqueness. Let  $\psi \in C^2(]0, T[)$  be a non negative test function with compact support in  $]0, T[$ . We write (17) with  $\phi = \psi \rho_\delta$  and take the limit as  $\delta$  tends to zero. Thus, by using the same arguments as previously, we have, for all real  $k$ ,

$$(20) \quad (f(\gamma u, x, t) - f(k, x, t)) \cdot n (\operatorname{sg}(\gamma u - k) + \operatorname{sg}(k)) \geq 0$$

almost everywhere on  $\Gamma \times ]0, T[$ . Inequality (20) is reduced to (16) by taking  $k$  into the interval  $I(\gamma u, 0)$ , and is obviously satisfied otherwise.

We now introduce two functions  $u$  and  $v$ , solutions of problem (1), (2), (16). By the same method as S.N. KRUSKOV [3], for all test function  $\phi$  equal to zero on  $\Gamma \times ]0, T[$ , we obtain

$$(21) \quad \left\{ \int_{\Omega} \int_0^T \left\{ |u-v| \frac{\partial \phi}{\partial t} + \operatorname{sg}(u-v) (f(u, x, t) - f(v, x, t)) \cdot \operatorname{grad} \phi \right. \right. \\ \left. \left. - \operatorname{sg}(u-v) \left[ \sum_i \frac{\partial f_i}{\partial x_i}(v, x, t) - \frac{\partial f_i}{\partial x_i}(u, x, t) + g(u, x, t) - g(v, x, t) \right] \right\} dx dt \right\} = 0.$$

As test function  $\phi$ , we take the following

$$(x, t) = \psi(t)(1 - \rho_\delta(x)) \quad \text{for } (x, t) \in \Omega \times ]0, T[,$$

where  $\psi$  and  $\rho_\delta$  are the same functions as above, and we introduce it into (21) to get at the limit, when  $\delta$  tends to zero,



$$(22) \quad \left\{ \begin{array}{l} \int_{\Omega} \int_0^T \left\{ |u-v| \psi'(t) - \text{sg}(u-v) \sum_{i=1}^p \left( \frac{\partial f_i}{\partial x_i}(v, x, t) - \frac{\partial f_i}{\partial x_i}(u, x, t) \right) \right. \\ \quad \left. + g(u, x, t) - g(v, x, t) \right\} \psi(t) dx dt \geq \\ \geq \int_{\Omega} \int_0^T \text{sg}(\gamma u - \gamma v) (f(\gamma u, s, t) - f(\gamma v, s, t)) \cdot n \cdot \psi(t) ds dt. \end{array} \right.$$

To determine the sign of the right-hand side of (22), we define, for all  $(x, t) \in \Gamma \times ]0, T[$ , a real  $k(x, t)$  by

$$k(x, t) = \begin{cases} \gamma u(x, t) & \text{if } \gamma u \in I(0, \gamma v), \\ 0 & \text{if } 0 \in I(\gamma v, \gamma u), \\ \gamma v(x, t) & \text{if } \gamma v \in I(\gamma u, 0), \end{cases}$$

which allows us to write

$$\begin{aligned} \text{sg}(\gamma u - \gamma v) (f(\gamma u, x, t) - f(\gamma v, x, t)) &= \text{sg}(\gamma u - k) (f(\gamma u, x, t) - f(k, x, t)) \\ &\quad + \text{sg}(\gamma v - k) (f(\gamma v, x, t) - f(k, x, t)). \end{aligned}$$

Now, by (20), we find that the right-hand side of (22) is non negative.

For an interval  $[t_0, t_1] \subset ]0, T[$ , we can take in (22) a test function  $\psi$  which approaches the characteristic function of this interval and from (22), we obtain at the limit

$$\left\{ \begin{array}{l} |u(\cdot, t_1) - v(\cdot, t_1)|_{L^1(\Omega)} \leq |u(\cdot, t_0) - v(\cdot, t_0)|_{L^1(\Omega)} \\ \quad + c_0 \int_{t_0}^{t_1} |u(\cdot, t) - v(\cdot, t)|_{L^1(\Omega)} dt, \end{array} \right.$$

where  $c_0$  is the sum of the Lipschitz constants with respect to the variable  $u$  of  $g$  and  $\frac{\partial f_i}{\partial x_i}$  for all  $i \in \{1, \dots, p\}$ . From Gronwall's Lemma, we conclude that

$$(23) \quad |u(\cdot, t_1) - v(\cdot, t_1)|_{L^1(\Omega)} \leq |u(\cdot, t_0) - v(\cdot, t_0)|_{L^1(\Omega)} e^{c_0(t_1 - t_0)},$$

which shows that the semi-group associated with equation (1) and boundary conditions (16) is Lipschitz continuous in the  $L^1$ -norm, and is a contraction for  $c_0 = 0$ . Theorem 2 is then proved since uniqueness can be immediately deduced from (23) as  $t_0$  tends to zero. ■

Approximations of problem (1), (2), (16) by a difference scheme may be proved to be convergent by using the same compactness argument as in Theorem 1. Paper [6] deals with the approximation by S. GODUNOV's scheme for the one-dimensional problem. The same techniques are efficient for other schemes, even in several variable cases.

Some of our results were presented in [7].

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